Signed Pell caps and free scale carriers for Diophantine consistency

Research notes for ProveIt

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Abstract

This packet develops explicit Diophantine consistency constructions across several size criteria. A signed pooled Pell cap compresses the direct-calculus quartic to 352 unrestricted integer unknowns. A free-scale carrier lowers the fifteen-integer degree to 44. Main-root saturation combined with the free-scale argument gives an eleven-integer polynomial of degree 352. Polynomial radical centering gives ten-positive-variable equations of degrees 253 and 252. A separate single-product saturation reduces both degree and cost at two low-operation eleven-integer endpoints.

The supplied ordinary proofs are reductions relative to pinned, retained compiler and refutation calibrations. They do not prove consistency, global optimality or a new end-to-end formal-kernel theorem. The accompanying archive contains literal equations and DAGs, complete coefficient payloads, proofs and independent reviews, source provenance, and offline normal and optimized Python replay with corruption controls.

Construction	Domain	Variables	Degree	Other size measure
Signed cap, odd quotient	$\mathbb{Z}$	352	4	219 rows; 9,355 terms
Signed cap, guarded quotient	$\mathbb{Z}$	355	4	220 rows; 9,370 terms
Free-scale first-root carrier	$\mathbb{Z}$	15	44	111 arithmetic gates
Free-scale main saturation	$\mathbb{Z}$	11	352	132 arithmetic gates
Main-and-guard product, native	$\mathbb{Z}$	11	924	109 arithmetic gates
Main-and-guard product, short	$\mathbb{Z}$	11	840	110 arithmetic gates
First reciprocal correction	$\mathbb{Z}_{>0}$	10	253	102 arithmetic gates
Second reciprocal correction	$\mathbb{Z}_{>0}$	10	252	104 arithmetic gates

The two direct quartics have coefficient height 15,020,718. Their variable saving is within the inspected small-coefficient direct-calculus family, not a fewest-variable quartic claim across native coefficient architectures. Most compact endpoints are degree–cost tradeoffs: several lower-cost predecessors remain nondominated. The two single-product endpoints improve both metrics relative to the inspected 966/111 and 882/112 pairs.

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1 Scope and the exact logical claim

The objective is an explicit fixed polynomial P with integer coefficients such that existence of a zero is equivalent to existence of a finite syntactic refutation of ZFC. For the domain D stated for each construction, the corresponding ordinary arithmetic theorem gives

$$\mathsf{ZFC} \vdash \left(\neg\exists \mathbf{x} \in D^n \; P(\mathbf{x}) = 0\right) \longleftrightarrow \mathrm{Con}(\mathsf{ZFC}).$$

This is an equivalence, not a proof of either side. No consistency assumption is used to establish the reductions. The new transformations are ordinary unbounded proofs relative to explicitly retained source calibrations; the packet does not supply a new end-to-end Lean, Rocq or encoded object-language ZFC certificate. The distinction concerns substantive inherited calibration theorems, not merely a missing formal proof file.

The report compares three distinct domains: nonnegative integers, strictly positive integers and unrestricted integers. A count in one domain is not silently transferred to another. A displayed polynomial abbreviation is not an extra unknown. Intermediate arithmetic DAG registers denote substitutions into the final polynomial, not existentially quantified values. Proof-only Pell sequences, divisions and radicals are not operations in the supplied division-free polynomial graphs.

Operation counts mean live binary additions, subtractions and multiplications. Squaring and multiplication by any fixed integer each count as one multiplication. Constant payload size and circuit cost are separate metrics; the compact native coefficient equations use enormous fixed numerals, whereas the direct calculus quartics have fully expanded, much smaller coefficients. There is no single ordering that makes one construction uniformly simplest.

For Pell notation, for $b \geq 2$ and $m \geq 0$ write

$$\chi_b(m) + \psi_b(m)\sqrt{b^2-1} = (b + \sqrt{b^2-1})^m.$$

Different source sections also call these coordinates $X_m(b), Y_m(b)$. They are used only in the mathematical proofs.

2 The pinned comparison frontier

The comparison snapshot is commit `f4d8db99e060`, the inspected main revision at 23:45:29 UTC on 24 October 2026. It includes eight incoming archives and is newer than the preceding release's comparison pin. In particular, the repository's convenient canonical display is not the entire current frontier.

A final head check at 00:05 UTC on 25 October found commit `12a6f5a0376b`, dated 00:01:49 UTC, accepting the complete 442-integer unified-binomial quartic after ordinary source and literal replay review. It is therefore an accepted comparator, not merely an incoming proposal. No newer competing metric was identified. This acceptance does not change the signed-cap result's inherited 489-parent calibration boundary. The corresponding review and a concise commit receipt are retained under `baseline/late/`; historical source pins remain stable even though the repository reorganized incoming archives into numbered sources.

Direct construction	Variables	Rows	Degree	Terms	Height
Shared Pell cap, nonnegative	261	194	4	31,679	85,566,955
Shared Pell cap, sparse nonnegative	292	225	4	6,140	83,933,334
Even indices, unrestricted	442	290	4	4,652	128,446
Weak carriers, unrestricted	478	305	4	5,484	511,326
Cofinal witnesses, unrestricted	508	308	4	6,545	996,521

The 352-variable signed-cap endpoint improves the count within the inspected small-coefficient direct-calculus integer family, but it has greater support and coefficient height than the 442-variable point. It is emphatically not the fewest-variable quartic overall: native fixed-coefficient architectures already have much smaller variable counts and extremely large constants. The nonnegative 261 point is not an unrestricted-integer point.

The inspected eleven-integer circuit frontier includes operation/degree pairs $(111, 966)$, $(112, 882)$, $(118, 488)$, $(119, 464)$, $(124, 456)$ and $(125, 416)$. These are the appropriate preceding comparisons, not the older local packet alone. The separate height-one construction already has all nonzero coefficients equal to ± 1; its 7431 variables address a different objective. No new height lower bound is sought here.

For ten positive variables the incorporated gap-fusion family already contains operation/degree pairs $(89, 288)$, $(91, 283)$, $(92, 278)$, $(94, 262)$, $(96, 258)$ and $(97, 254)$. Their overlap with the preceding Paired Guards and Centered Carriers release is already acknowledged upstream. They are not counted as new round-16 advances. The exact one-gate rebracketing

$$(q^3 q)(q-1) = q^3(q(q-1))$$

is also part of the comparison baseline, not a new mathematical reduction.

3 Coefficients and inherited calibration

The integer compact branches use the complete original-label source34 tuple, including $\beta = 2^{1064921} - 1$, K_0, M_C, the already shifted M_F, and the two cone coefficients $2^{1113}, 2^{666104}$. The ten-positive branch uses the complete original compact44 tuple with $\beta = 2^{590069} - 1$. All coefficients are fixed explicit integers supplied as compressed minimal unsigned little-endian byte strings. Their bindings record hashes, conventions and bit lengths. The historical recipes and calibration proofs are retained in the exact preceding release. New indexed or compact controller tuples cannot be substituted coefficient by coefficient.

For the direct branch, the relevant inherited chain is the fixed finite C222 calculus, square-shell source construction, factorial and reciprocal-CRT arithmetic, derived-label range, and hybrid/Horn-flow extraction. The new proof changes the indexed-Pell layer and then applies checked integral monic projection. The numerical polynomial compiler does not itself prove the calculus-to-ZFC correspondence.

For the compact branches, the new equations reduce to the retained normalized Pell and native compiler interfaces. The zero-existence chain is

$$\exists \text{ polynomial zero} \iff \exists \text{ accepted finite source certificate} \iff \exists \text{ finite syntactic ZFC refutation.}$$

The current independent checks address the newly changed algebra, literal graphs, degree attainment, source payloads and proof obligations stated in their reviews. They do not freshly rebuild the complete transitive compiler, Pell-calibration and formal-system chain. Checksums authenticate the retained documents; they do not turn their assertions into proofs.

4 A signed pooled Pell cap: 352 unrestricted integers

The new direct-calculus polynomial is the sum of 219 squared quadratic residuals in 352 unrestricted integer variables. Its complete expansion has 9,355 monomials and coefficient height 15,020,718. Relative to the newly reviewed 442-integer direct quartic, this saves 90 coordinates (20.4%), while increasing support and coefficient height. It is not a globally smallest-variable quartic; native large-coefficient architectures are much smaller.

The key new principle recovers signs and indices together. Suppose $A-1>9W^3$, $1\le k_i<W$, and $m\le W$. Retain only the signed principal norms $x_i^2-(A^2-1)y_i^2=1$. Certify once $V=Y_{2W}(A)$ and impose

$$V=2W+1+s+\sum_i y_i^2,\quad s\ge 0,\qquad \sum_i (y_i-k_i)^2=(A-1)Q.$$

Classification gives $y_i=\epsilon_i Y_{j_i}(A)$. The squared cap implies $j_i<2W$. Reducing the pooled congruence modulo $A-1$ therefore gives a multiple $E=\sum_i(\epsilon_i j_i-k_i)^2$ with $0\le E<9W^3<A-1$. Hence $E=0$, and every ordinate is positive and has the intended index. Completeness follows from $Y_{2W}=2X_WY_W>2(A-1)Y_W^2$, which dominates all the squared ordinates. The old signed-power inverse-branch argument separately orients each abscissa; the cap alone asserts only its absolute value.

A single finite indexed-Pell certificate at $2W$ supplies V. Its auxiliary ordinate can be constrained by $J=V^2(2e+1)$ with signed e, without a sign guard. Indeed the canonical V is even and its abscissa U is odd. Choosing $J=Y_{2WV}(A)=VY_V(U)$ gives

$$J/V^2=U^{V-1}+(A^2-1)VT,$$

an odd integer. This saves three additional guard witnesses relative to the 355-integer sibling, which is also supplied.

All variables, coefficients, residuals, and monic restorations are explicit. The checker replays 229 projections, verifies the 41-atom frame and all 19 retained power requests, and expands the complete quartic. The ordinary proof inherits the frozen parent's C222/refutation and ZFC calibration; it is not a new kernel proof or an assumption of consistency. The squared signed cap extends, and credits, the incoming natural-domain pooled cap.

5 Complete ordinary proof of the signed-cap construction

5.1 Literal polynomial and logical scope

The preferred polynomial is the fixed polynomial

$$P_{352}(z_0,\dots,z_{351})=\sum_{r=0}^{218} R_r(z_0,\dots,z_{351})^2\quad\text{in }\mathbb{Z}[z_0,\dots,z_{351}].$$

Here the ordered list of all 219 quadratic residuals R_r, including its complete ordered variable list and every integer coefficient, is the literal file `alternative/odd_cap/equation.json`. A term `[c,[i,j,...]]` means $cz_iz_j\cdots$, and an empty index list means a constant. The complete collected coefficient list defining the same polynomial is `alternative/odd_cap/expanded_polynomial.json`. Its SHA-256 is

`a4b6e1f295015967ebf3cfe8f5bf397b1e1a9987b1d620fcd940f4cef8f1ce84.`

Thus the equation is an ordinary fixed polynomial equation, not a formula with powers, factorials, Pell functions, or unbounded lists left as primitive operations. Every unknown ranges over all of $\mathbb{Z}$. The exact degree is four; there are 9,355 collected monomials and the maximum absolute collected coefficient is 15,020,718. The simpler 355-variable sibling is defined analogously by the 220 residuals in `alternative/signed_cap/equation.json`. It has 9,370 monomials and the same degree and coefficient height.

The proof below establishes the new arithmetic replacement in both directions. Composed with the frozen parent's ordinary direct-arithmetic, C222/refutation, and ZFC calibration, it gives

$$\mathrm{ZFC}\vdash\big(\exists z\in\mathbb{Z}^{352}\ P_{352}(z)=0\big)\longleftrightarrow\big(\text{there exists a finite ZFC refutation}\big).$$

Consequently its no-zero sentence is equivalent in ZFC to Con(ZFC). This is an ordinary mathematical proof with the stated inherited calibration, not a newly completed Lean or Rocq kernel proof or a newly printed object-language ZFC derivation. Neither consistency nor inconsistency is assumed. The construction does not simulate K215 or instantiate Jones's universal polynomial. Variable-count comparisons here concern the direct-calculus, relatively small-coefficient quartic architecture; other architectures have much smaller variable counts and enormous fixed coefficients.

5.2 The signed squared-cap theorem

For $A \geq 2$ define the integer Pell sequences by

$$X_0(A) = 1, \qquad Y_0(A) = 0,$$

$$X_{j+1}(A) = AX_j(A) + (A^2-1)Y_j(A),$$
$$Y_{j+1}(A) = X_j(A) + AY_j(A).$$

Assume the family is nonempty, so $m \geq 1$. Suppose

$$W \geq 4, \qquad A - 1 > 9W^3, \qquad 1 \leq k_i < W \ (1 \leq i \leq m), \qquad m \leq W.$$

Let x_i, y_i be unrestricted integers satisfying

$$x_i^2 - (A^2-1)y_i^2 = 1.$$

Certify once that $V = Y_{2W}(A)$, and impose

$$V = 2W + 1 + s + \sum_{i=1}^{m} y_i^2, \qquad s \geq 0, \tag{1}$$

$$\sum_{i=1}^{m} (y_i - k_i)^2 = (A-1)Q, \qquad Q \in \mathbb{Z}. \tag{2}$$

Then

$$y_i = Y_{k_i}(A) > 0, \qquad |x_i| = X_{k_i}(A) \quad (1 \leq i \leq m).$$

Conversely the canonical positive ordinates and either signs of their abscissae extend to solutions of (1)–(2).

Soundness. Integer Pell classification gives indices $j_i \geq 0$ and signs $\epsilon_i \in \{1, -1\}$ such that

$$|x_i| = X_{j_i}(A), \qquad y_i = \epsilon_i Y_{j_i}(A).$$

Either sign may be used when $j_i = 0$. Equation (1) gives $y_i^2 < V$. Since $V > 1$, it follows that $|y_i| < V$, and strict monotonicity of $Y_j(A)$ gives $j_i < 2W$. In particular the endpoint $j_i = 2W$ is excluded.

Modulo $A - 1$, the recurrence gives $Y_j(A) \equiv j$. Therefore (2) implies

$$A - 1 \ \mid \ E := \sum_{i=1}^{m} (\epsilon_i j_i - k_i)^2.$$

Every absolute difference is less than $3W$, so

$$0 \leq E < 9mW^2 \leq 9W^3 < A - 1.$$

Thus $E = 0$. Each signed index $\epsilon_i j_i$ equals the positive integer k_i, which forces $\epsilon_i = 1$ and $j_i = k_i$. Neither a negative ordinate nor a zero-index solution survives. This argument deliberately does not orient the sign of x_i; that is a separate consequence of the retained power rows below.

Constructive completeness. Set $y_i = Y_{k_i}(A)$. Every difference $y_i - k_i$ is divisible by $A-1$, so the quotient Q in (2) is an integer, indeed nonnegative. Put $T = Y_W(A) \geq 1$. The norm and positivity imply $X_W(A) > (A-1)T$. The double-index identity therefore gives

$$V = Y_{2W}(A) = 2X_W(A)T > 2(A-1)T^2.$$

Since each $k_i < W$, one has $y_i < T$ and $\sum_i y_i^2 \leq mT^2$. Consequently

$$V - \sum_i y_i^2 > (2A - 2 - m)T^2 > 2W + 1,$$

using $A - 1 > 9W^3$ and $m \leq W$. Hence the explicit restoration

$$s = V - 2W - 1 - \sum_i y_i^2$$

is nonnegative. This proves completeness without extrapolating from finite examples or imposing an additional asymptotic assumption.

5.3 A single ordinary indexed-Pell certificate supplies the cap

Initially consider the simpler sibling. All letters below are integers, with only e expressly constrained to be nonnegative:

$$U^2 - (A^2-1)V^2 = 1, \qquad H^2 - (A^2-1)J^2 = 1, \tag{3}$$

$$S^2 - (B^2-1)R^2 = 1, \qquad B = 1 + 4Vd, \tag{4}$$

$$J = V^2(e+1), \qquad e \geq 0, \tag{5}$$

$$B = A + Hf, \qquad S = U + Hg, \tag{6}$$

$$R = 2W + 4Vh. \tag{7}$$

The aggregate cap gives $V \geq 2W + 1 > 0$. Thus $J \geq V^2 > 1$. Auxiliary Pell classification implies $|H| \geq A \geq 2$, in fact $|H| \geq 2A^2 - 1$.

No positivity guard on B or d is needed. If $d = 0$, then $B = 1$ and (6) would give $H \mid A - 1$, contrary to $|H| \geq A$. If $d \neq 0$, then $B = 1 + 4Vd$ gives $|B| \geq 4V - 1 \geq 3$. Thus $|B| \geq 2$. The coordinates $d, B, f, S, R, g, h, H, U$ may all remain unrestricted.

The inherited signed indexed-Pell criterion applies with requested index $2W$, principal ordinate V, auxiliary ordinate J, and moduli $|H|$ and $4V$. Its hypotheses

$$1 \leq 2W \leq V, \qquad V^2 \mid J,$$

the three norm equations, and its four congruences are precisely the rows above and the aggregate cap. The conclusion is

$$|U| = X_{2W}(A), \qquad V = Y_{2W}(A).$$

For completeness, the signed extensions used in this application are spelled out rather than silently assumed.

For negative B, define $X_j(B), Y_j(B)$ by the same polynomial recurrence. For $j > 0$,

$$X_j(B) = (-1)^j X_j(|B|), \qquad Y_j(B) = (-1)^{j-1} Y_j(|B|),$$

while $X_0(B) = 1$, $Y_0(B) = 0$. Thus the classification at $|B|$ permits independently signed coordinates at the signed base B. Polynomial base-congruence transfer and $Y_j(B) \equiv j \pmod{B-1}$ continue to hold. The usual residue-class proof consequently applies for $|B| \geq 2$. A signed principal U may be replaced by

$|U|$ while multiplying S by the same sign. Its norm and the congruence $S \equiv U \pmod{H}$ are preserved. The sign of H changes no divisibility modulus.

Here is the precise short-interval step. Classification gives a principal index $i > 0$, an auxiliary index $n > 0$, and a comparison index $r \geq 0$. Squared divisibility $V^2 \mid J$ gives $V \mid n$, and the size comparison gives $i \leq n$. The abscissa congruence, with the independent sign of S and the half-period sign-change identity allowed, gives

$$r \equiv i \text{ or } -i \pmod{2n}.$$

The ordinate congruence gives

$$r \equiv 2W \text{ or } -2W \pmod{4V}.$$

It follows that $i \equiv 2W$ or $-2W \pmod{2V}$. Since $1 \leq i \leq V$ and $1 \leq 2W \leq V$, this forces $i = 2W$, including the endpoint case. This is the inherited indexed-Pell argument applied to the stated signed domains.

For the simpler sibling's completeness, choose the canonical pair at $2W$. The existing indexed-Pell completeness theorem supplies H, J, B, S, R and progression witnesses with $B > 0$ and $e \geq 0$. They also satisfy the weaker unrestricted domains here. Only one ordinary indexed certificate, not a stronger simultaneous theorem, is required for this cap. An explicit strengthened construction for the preferred sibling is provided below.

5.4 The scale and seed premises are established before any powers

The retained parent frame is the exact polynomial identity

$$W = 2 + \sum_{a \in \mathcal{A}} a^2,$$

where $\mathcal{A}$ is the set of distinct actual bases, exponents, and outputs of the nineteen retained power requests. The independent checker verifies the fully expanded identity against the literal 41-atom inventory. Consequently every such atom has absolute value less than W. Since $t = t_0 + 16$ with $t_0 \geq 0$, and t is one of these atoms, $W \geq 258$. In particular the sixteen independent seeds satisfy $m = 16 \leq W$.

The retained scale certificate, with $A, z \geq 0$, is

$$A^2 - \big((W+1)^2 - 1\big)\big(W(z+1)\big)^2 = 1.$$

Its ordinate is positive. Classification at base $W+1$ and $Y_\ell(W+1) \equiv \ell \pmod{W}$ imply a positive index ℓ divisible by W. Therefore

$$A \geq X_W(W+1) \geq (W+1)^W.$$

For every $W \geq 4$,

$$(W+1)^W - 1 > 9W^3.$$

At $W = 4$ this is $624 > 576$. The ratio of successive exponential bounds is greater than $W + 2$, whereas the cubic ratio is at most $(5/4)^3$, so the inequality persists by induction. It follows that $A - 1 > 9W^3$ without interpreting any requested power.

All sixteen seed exponents are already positive from retained semantic guards and literal source rows. They are t, $K = 2R$, $\rho + t$, $F^2 + t$, the nine positive encoded representatives, $y + 1$, $x + 1$, and the positive derived-label expression. The presemantic definitions are

$$\rho = F^2 t^2, \qquad y = F^2 + r_0 + 1, \qquad R = 2 + t + x + y + F^2, \qquad x, r_0, t_0 \geq 0.$$

The sign of F is not yet assumed. The derived-label expression has a positive constant term and nonnegative constructor inputs. The fresh literal lower-bound certificate proves every seed from retained guards and the actual source rows; it uses neither a removed seed-gap sign nor any conclusion of the cap theorem. Thus the signed cap theorem applies before any factorial, binomial, CRT range, or power output has been interpreted.

5.5 Exact changed interface and quadratic implementation

The frozen source is the square-anchored 489-integer direct construction before its monic projection. For each of sixteen independent seeds the principal norm and all power-congruence rows are retained. The deleted rows are exactly its index-at-most-Y row, comparison norm, comparison-abscissa congruence and comparison-ordinate congruence; the old shared auxiliary norm, shared comparison-base congruence and positive-product row are also deleted. The sixteen scalar guards on $Y - k$ and two old shared auxiliary scalar guards are removed. The precise original row numbers and every deleted literal are recorded in `cap_certificate.json`.

No semantic sign, digit-pair guard, factorial ceiling, range row, or Horn-flow source row is deleted. Unused arithmetic gates of the old certificates remain temporarily as polynomial definitions, then disappear under checked integral monic projection. They are not premises of the new proof.

Every nonnegative slack is represented by the ternary form

$$\mathcal{N}(a,b,c) = a^2 + b^2 + c(c+1), \qquad a,b,c \in \mathbb{Z}.$$

This takes only nonnegative values and represents every nonnegative integer. Indeed Legendre's three-square theorem applies to $4n+1$; exactly one of the three representing coordinates is odd, and a permutation followed by division by four gives this form.

To remove any ambiguity about degree, introduce the four explicit wires

$$a_V = AV, \qquad a_J = AJ, \qquad b_R = BR, \qquad v_2 = V^2.$$

For the preferred odd-quotient construction, the entire new cap interface consists of the following quadratic residuals:

$$\begin{array}{llll}
a_V - AV, & a_J - AJ, & b_R - BR, & v_2 - V^2, \\
U^2 + V^2 - a_V^2 - 1, & H^2 + J^2 - a_J^2 - 1, & S^2 + R^2 - b_R^2 - 1, & \\
B - 1 - 4Vd, & J - v_2 - 2v_2 e, & B - A - Hf, & \\
S - U - Hg, & R - 2W - 4Vh, & & \\
V - 2W - 1 - s - \displaystyle\sum_i y_i^2, & & & \\
\displaystyle\sum_i (y_i - k_i)^2 - AQ + Q, & & & \\
s - a^2 - b^2 - c^2 - c. & & &
\end{array}$$

Here a, b, c are the three fresh witnesses for the aggregate slack, not the comparison base B. Every k_i is affine in already existing source wires, so its squared contribution is quadratic without extra summand variables. In the simpler sibling, replace $J - v_2 - 2v_2 e$ by $J - v_2 - v_2 e$ and add $e - a_e^2 - b_e^2 - c_e^2 - c_e$. All variable names, exact indices and fresh guard witnesses are fixed in the supplied finite certificate. The projection may subsequently remove some of these wires; their restorations are supplied explicitly.

5.6 All nineteen powers and their abscissa orientations

After the cap theorem, each seed has

$$Y = Y_k(A) > 0, \qquad X = X_k(A) \text{ or } -X_k(A).$$

For a retained request at positive base n and index k, the unchanged row is

$$X + (n - A)Y = m + T_n u, \qquad T_n = 2An - n^2 - 1.$$

The frame supplies $1 \le n, k < W$ and $|m| < W$; its scale gives $T_n > W^W + W$. If $X > 0$, standard Pell substitution gives $m \equiv n^k \pmod{T_n}$, and the strict interval bound yields $m = n^k$.

If $X < 0$, multiply the conjugate expressions and use the norm. The result is

$$mn^k \equiv -1 \pmod{T_n}.$$

Since $|mn^k + 1| < T_n$, this forces the integer identity $mn^k = -1$. For $n \ge 2$ it is impossible. At $n = 1$ the sole inverse branch is $m = -1$. Thus a request with base at least two orients its shared principal pair. This reasoning uses no nonnegative quotient or output assumption: all such quotients remain signed.

The complete noncircular recovery order is as follows.

1. The retained factorial remainder and slack give $D \ge 1$. The retained exact-binomial output slack gives $q = D + o + 1 \ge 2$. The request q^t orients the t pair. The request $(\rho + 1)^t$ sharing that pair therefore recovers its true value $V_0 \ge 1$, even if $\rho + 1$ initially equals one. The unchanged ceiling identity $DF = V_0 +$ factorial slack now gives $F \ge 1$, and consequently $V_0 \ge 2$.

2. The retained $B \ge 0$ guard and $C + F^2 + 1 = Bi_q$, with $C \ge 1$, imply $B > 0$. The request at index K for B^K gives either $N = B^K > 0$ or the sole inverse case $B = 1$, $N = -1$. In either case the second base $N + V_0 + 1$ is at least two. That request orients the K pair, excludes the inverse branch, and recovers the common radix $q = N(N + V_0 + 1)^K$.

3. Every other independent seed has a request at base q or $q + 1$, both at least two. Those requests orient all remaining principal pairs. The $t + 1$ pair is obtained by the retained exact Pell addition from the already oriented t pair, not by an unchecked extra seed.

In this paragraph B is the retained semantic binomial output; it is distinct from the private comparison-base coordinate B of the cap certificate, whose scope was local to that certificate. The literal source assigns distinct variable indices to them.

For completeness, the entire request inventory in original output labels is:

- `pow0`: $(\rho + 1)^t$; `pow4`: q^t.
- `pow1`: B^K; `pow2`: $(N + V_0 + 1)^K$.
- `pow3`: $(q + 1)^{\rho + t}$; `pow5`: $(q + 1)^{F^2 + t}$.
- `pow6`, `pow8`, `pow9`, `pow10`, `pow11`, `pow13`, `pow14`, `pow16`, `pow17`: $(q + 1)$ raised to the nine shifted encoded representatives.
- `pow12`: q^{y+1}; `pow15`: q^{x+1}.
- `pow18`: $(q + 1)$ raised to the positive derived-label expression.
- `pow7`: q^{t+1}, using the derived addition pair.

This is nineteen requests at sixteen independent indices and one derived index. The literal audit checks retention of every row, its exact exponent binding, and coverage of every independent seed.

5.7 Composition with the unchanged direct arithmetic

Once the powers are recovered, the twelve paired guards of the parent restore all digit remainder and slack signs from their nonnegative sums $q^k - 1$. The parent's ordered factorial proof then gives $F = t!$, identifies the reciprocal binomial product, and validates every CRT range digit. Its amplified reciprocal CRT argument recovers each bounded local C222/Horn-flow row and the derived label bound. The

multiplicity-flow argument extracts a finite refutation, to which the ordinary C222/ZFC calibration applies. These retained arguments are mathematical dependencies, not conclusions inferred from finite regression tests.

Conversely, start from a finite refutation and choose the semantic data exactly as in the parent's constructive completeness: the horizon, factorial, Horn accumulators, beta-CRT data, reciprocal product, powers, representatives and digit witnesses. These data determine W. The retained scale construction supplies A. Choose every seed canonically at A, and derive the $t+1$ pair by addition. The old positive-power completeness supplies all retained power quotients. The signed-cap completeness proved above supplies the aggregate slack and pooled quotient, and the single cap-index certificate supplies its private witnesses. Add the required ternary slack representations and retain all semantic guards. Assign unused old witness coordinates arbitrarily, then compute their remaining gate definitions. Every new raw residual vanishes.

This reconstruction may replace the private old Pell witnesses. It claims existence-preserving equivalence, not same-tuple equality of discarded coordinates. The Horn factorization is the parent's exact polynomial identity, and every subsequent elimination has an explicitly checked integral monic restoration. Thus the final finite polynomial has exactly the stated existence status in both directions.

5.8 Odd auxiliary quotients: the preferred 352-variable form

Replace only (5) by

$$J = V^2(2e+1), \qquad e \in \mathbb{Z}.$$

The single sign predicate for e and its three witnesses are removed. The remaining interface is unchanged. The actual emitted polynomial, not merely the local count, has 352 variables and 219 quadratic residuals.

Soundness with a signed auxiliary ordinate. For integer e, the number $2e+1$ is nonzero and has absolute value at least one. Hence $|J| \geq V^2 > 1$. The auxiliary norm still gives $|H| \geq A$. Every use of J in the signed indexed-Pell theorem is through its absolute-value classification or the divisibility $V^2 \mid J$; its sign is not used. The cap proof therefore remains valid with $|J|$ in place of J in the lower bound, and no other premise changes.

Constructive odd quotient and the full private certificate. Let $r = 2W$, and take the canonical cap pair

$$U = X_r(A), \qquad V = Y_r(A).$$

The doubling identities make V even and U odd. Choose

$$H = X_{rV}(A), \qquad J = Y_{rV}(A) = VY_V(U),$$

where the last equality is the Pell composition identity. Polynomial expansion of $Y_V(U)$ gives an integer T with

$$Y_V(U) = VU^{V-1} + (U^2-1)T.$$

As $U^2 - 1 = (A^2-1)V^2$, this yields

$$\frac{J}{V^2} = \frac{Y_V(U)}{V} = U^{V-1} + (A^2-1)VT.$$

The right side is a positive odd integer: U is odd and V is even. Thus

$$e = \frac{J/V^2 - 1}{2} \in \mathbb{Z}$$

provides the required auxiliary row.

Moreover $H = X_V(U)$ is odd, since U is odd and $U^2 - 1$ is even. The norm gives $\gcd(H, J) = 1$; because $V \mid J$, it follows that $\gcd(H, 4V) = 1$. The ordinary Chinese remainder theorem therefore supplies arbitrarily large positive B such that

$$B \equiv A \pmod{H}, \qquad B \equiv 1 \pmod{4V}.$$

Choose one and set $S = X_r(B)$, $R = Y_r(B)$. Polynomial base transfer gives $S \equiv U \pmod{H}$, and base-one reduction gives $R \equiv r \pmod{4V}$. The corresponding integers d, f, g, h satisfy every remaining cap row. This is an explicit construction of the whole private certificate; no unproved odd-quotient version of an inherited completeness statement has been assumed.

The saving is precisely three coordinates and one quadratic residual. Only the aggregate slack needs a ternary nonnegative predicate. The simpler sibling remains independently supplied.

5.9 Reproduction, finite checks, and attribution

The complete local ordinary proof is `alternative/PROOF.md`. To rebuild the preferred endpoint, run `python build_signed_cap.py NEW -odd-auxiliary`; then run `python check_signed_cap.py NEW`. Omitting the flag builds the simpler sibling. Both procedures also pass with Python optimization enabled. The producer uses only standard-library arithmetic and frozen reviewed polynomial utilities; no remote state is modified.

The checkers verify the exact parent deletion boundary, strict new row and variable inventories, absence of the omitted guard, all sixteen presemantic index bounds, the 41-atom frame identity, the nineteen request bindings, every integral monic restoration, and the full collected quartic. The independent Horn audit checks 17 affine square roots, 11 product links, 18 literal rewrites, and the absence of any escaped factor. Each endpoint has 229 checked monic restorations. All four fresh builds, comprising two endpoints in normal and optimized modes, reproduce every producer artifact byte for byte. Altered-input, certificate, extra-row and coefficient controls are rejected. Small Pell examples and concrete weakened-hypothesis counterexamples supplement these checks; they do not decide consistency or replace any unbounded mathematical argument.

The natural pooled-cap principle is credited to the incoming `ConZFC_Shared_Pell_Bound_261_2026-10-24` packet. The new step is the squared aggregate at cap index $2W$, which recovers the signs and indices jointly and removes all individual principal-ordinate sign and index tests. The signed comparison-base extension, parent paired guards, square factorial anchor, Horn factorization, direct numerical calculus and ordinary ZFC calibration retain their original attribution and source pins. No claim of a globally minimal equation or literature-wide historical priority is made.

6 A free-scale carrier of degree 44

This endpoint uses fifteen unrestricted integer variables, has exact degree 44, and a division-free evaluator with 111 binary arithmetic gates (63 multiplications, 26 additions, 22 subtractions). It improves the prior degree 48 at a cost of three gates; it does not Pareto-dominate the retained 108-gate circuit. The same elementary projection factorization reduces that old evaluator to 107 gates, so the apples-to-apples evaluation cost is four gates. The new device is to release the scale variable, replace premature coprimality by a magnitude guard, and recover the exact scale only after an amplified whole-product projection.

6.1 Complete polynomial

The variables are

$$j, C, z, h, r, v, w, T, c, n, f, y, p_1, p_2, p_3 \in \mathbb{Z}.$$

The fixed numerals $\beta, K_0, M_C, M_F, A_c, B_c$ are exactly the six pinned source34 coefficient payloads; M_F is already shifted. All the following are polynomial abbreviations, not witnesses:

$$\begin{gathered}
J = 6j + 1, \quad q_m = \beta J, \quad q = q_m + 1, \quad W = 12w + 4, \\
X = qW, \quad Y = v, \quad F = (K_0 + W)C - zq_m, \\
R = (q(q - F) - C)(q^2 - 1) + (M_C + qM_F)J, \\
E = XY, \quad a = E + Y, \quad H = 4a + 3, \quad \Delta = a^2 + H, \\
k = R + 1 + hE, \quad \eta = c - Yk, \quad L = E(Yk), \\
I = (2r + 1)c^2, \quad Q = \Delta(f^2 - 1), \quad V = f^2(cT - R) - c, \\
M = 2q^3, \quad U = MH, \quad K_{\mathrm{lo}} = 2X^2 - 3Xc - 2, \\
Z = Un + MK_{\mathrm{lo}} + H(Y - q^3), \\
D_{\mathrm{err}} = (A_cF - q)^2 + (B_cC - q)^2, \\
B_0 = k^2(q^2 - D_{\mathrm{err}}) - 4q^2\eta^2, \\
B_{\mathrm{new}} = Y^2B_0 - k^2q^8, \qquad P_{\mathrm{os}} = 1 + p_1^2 + p_2^2 + p_3(p_3 + 1).
\end{gathered}$$

Define the residuals

$$\begin{aligned}
A &= (Z + Uk)^2 + 4U(LZ - U), \\
B &= f^2 - \Delta I^2 - 1, \\
C_{\mathrm{aux}} &= V^2 - (Q - 1)y(2V + y) - 1, \\
G &= B_{\mathrm{new}} - P_{\mathrm{os}}.
\end{aligned}$$

The evaluator uses the equivalent schedules

$$K_{\mathrm{lo}} = X(2X - 3c) - 2, \qquad B_{\mathrm{new}} = (Yk)^2(q^2 - D_{\mathrm{err}}) - q^2\big(4(Y\eta)^2 + (kq^3)^2\big),$$

reusing Yk and q^3. The sole equation is

$$\boxed{P = A^2 + B^2 + C_{\mathrm{aux}}^2 + G^2 = 0.}$$

6.2 Monicity before coprimality

The actual $\beta = 2^{1064921} - 1$ is odd and $1 \pmod 3$. Consequently q is nonzero, even, and $2 \pmod 3$. The polynomial H is odd and nonzero, so $U \neq 0$ on every assignment. At a zero, $t = Z/U \in \mathbb{Q}$ satisfies

$$t^2 + 2(2L + k)t + k^2 - 4 = 0.$$

Monicity makes t integral. Modulo 2, $t + k$ is even, so $s = (t + k)/2$ is integral and

$$s^2 + (2s - k)L - 1 = 0, \qquad (s + L)^2 = 1 + L(L + k).$$

The same divisibility $U \mid Z$ gives

$$M \mid H(Y - q^3), \qquad H \mid MK_{\mathrm{lo}}.$$

There is deliberately no early assertion that H and q are coprime. Since H is odd and q even, the first relation nevertheless gives Y even, hence Δ odd. If $K = X(X + 2ac) - 1$, then $2K - K_{\mathrm{lo}} = HXc$, so oddness of H gives

$$H \mid q^3K.$$

6.3 The guard replaces the early scale hypothesis

The polynomial P_{os} is strictly positive and represents every positive integer by the three-square theorem. Hence

$$Y^2 B_0 > k^2 q^8 > 0,$$

where $k \neq 0$ follows because $k = 0$ would make $B_{\mathrm{new}} \leq 0$. Since $B_0 \leq k^2 q^2$, this proves

$$B_0 > 0, \qquad |Y| > |q|^3.$$

The unchanged centered-budget argument gives common strict signs for $q, F, C, \alpha = q - F - C$, and

$$|\eta/k| < \tfrac{1}{2}, \qquad |k|(|Y| - \tfrac{1}{2}) < |c| < |k|(|Y| + \tfrac{1}{2}), \qquad \operatorname{sgn}(c) = \operatorname{sgn}(kY).$$

Put $q_* = |q|$, $x = |X|$, $m = |Y|$. The authentic packing estimates give

$$q_* \geq 16, \quad q_*(q_*^2 - 1) < R < q_*^4 - 2, \quad x \geq 4q_* \geq 64, \quad |E| > 4q_*^4 > 3R + 1.$$

These replace the old literal cubic-scale expression wherever it was used before projection. Its remaining parity consequence, Δ odd, was already proved without coprimality.

6.4 Signed Pell bootstrap: inherited statements and hypotheses

The unchanged strong and auxiliary residuals recover the missing main Pell ordinate by the predecessor's odd-index folding argument. For $B_p = |a + 2|$, $c_* = |c|$, and $I_* = |2r + 1|c_*^2$, they give

$$|f|^2 - (B_p^2 - 1)I_*^2 = 1, \qquad |f| > 2I_*.$$

With $S = \Delta I_*$ the auxiliary row is

$$S^2 V^2 - (S^2 - 1)(V + y)^2 = 1.$$

The auxiliary Pell index is odd. Its polynomial identity, reduced modulo $|f|$, gives an ordinate congruent to $\pm c_*$. Folding the index into the interval from 0 to the strong index makes the ordinate at most I_*. Since $I_* + c_* < |f|$, this congruence gives an actual main ordinate $c_* = \psi_{B_p}(p)$.

The first Pell classification gives $|k| = 2\psi_{|2XY^2+1|}(n_0)$. The inequality $|c| > |k|$, the parameter comparison, the congruence $k = R + 1 + hE$, and $|E| > 3R + 1$ give the same signed first-index lower bound as in the source. The unchanged strong-rank lemma and signed auxiliary step-down then give

$$p = R, \qquad 2n_0 = R + 1, \qquad R = 2s_0 + 1, \quad s_0 \geq 24.$$

The sign-uniform ratio estimate uses only these classifications, $x \geq 64$, $m \geq 6$, and $|c/k| < m + 1$. It proves

$$m > \frac{(x-3)^{s_0}}{2} - 1, \qquad |H| > x(x-3)^{s_0}.$$

These are precisely the inherited unbounded Pell steps, not conclusions of finite testing. Their complete statements and the verified hypotheses are detailed in `PROOF.md`, Sections 3–4, with pinned source texts. No main projection or power-of-two property has yet been used.

6.5 Amplified projection and late scale recovery

Let $\sigma = \operatorname{sgn}(c)$ and $x_0 = \sigma X$. The signed polynomial Pell projections give

$$2^R K \equiv (x_0 - 2^R)(2^R x_0 + 1) \pmod{H}.$$

Thus H divides q^3 times this whole product. We bound that larger product instead of canceling a possible nonunit modulo H. For $x \geq 64$ and $s_0 \geq 24$,

$$(x/32)^{s_0} \geq (x/32)^{24} \geq x^4,$$

since $x^{20} \geq 64^{20} = 32^{24}$. Therefore

$$|H| > x(x-3)^{s_0} \geq x(x/2)^{s_0} \geq 16^{s_0} x^5.$$

Since $q_* \leq x/4$ and $R = 2s_0 + 1$,

$$q_*^3 2^{2R+2} x^2 \leq \frac{x^3}{64} 16^{s_0+1} x^2 = \tfrac{1}{4} 16^{s_0} x^5 < |H|.$$

This strictly bounds the absolute value of the divisible whole product. The product is zero. Its factor $2^R x_0 + 1$ is odd and cannot vanish, so $x_0 = 2^R$. Because $X \equiv 2 \pmod 3$ and R is odd, $\sigma = 1$ and $X = 2^R$.

The zero whole product and the same Pell congruence now give $H \mid 2^R K$. Since H is odd, $H \mid K$ and hence $H \mid K_{\mathrm{lo}}$. Set $g = K_{\mathrm{lo}}/H \in \mathbb{Z}$. Substituting $K_{\mathrm{lo}} = Hg$ into $MH \mid Z$ and canceling the nonzero integer H gives

$$M \mid Y - q^3, \qquad v_{\mathrm{scale}} = (Y - q^3)/(2q^3) \in \mathbb{Z}.$$

This is cancellation in the integers, not cancellation of a nonunit modulo H. In particular, no CRT coprimality claim is needed. Thus the exact old cubic scale is recovered after, rather than before, the signed bootstrap. The surviving index sign gives $k > 0$; with $c > 0$, this also gives $Y > 0$. The inherited centered half-binomial extraction now applies without change.

6.6 Explicit maps and canonical completeness

From any new zero construct a degree-48 predecessor zero as follows. Retain $j, C, z, h, r, w, T, c, f, y$, and set

$$v_{\mathrm{old}} = Y/q, \qquad n_{\mathrm{old}} = t - g - v_{\mathrm{scale}}.$$

The old carrier's quotient is exactly

$$\frac{Z_{\mathrm{old}}}{U_{\mathrm{old}}} = n_{\mathrm{old}} + g + \frac{v_{\mathrm{old}} - q^2}{2q^2} = t.$$

Its strong and auxiliary rows are unchanged. Represent the positive integer B_0 by its three Pos coordinates. This restores a literal old zero, so the old native/compiler/refutation theorem applies.

Conversely start with a finite syntactic refutation of ZFC and use the same authenticated canonical compiler family. It supplies an old zero with $q > 0$, $X = 2^R$, $s_0 \geq 24$, the exact cubic scale, and

$$D_{\mathrm{err}} \leq q^2/2, \quad 0 < \eta/k < 1/4, \quad B_0 > q^2 k^2/4, \quad Y \geq X^{s_0}/2 > 2q^3.$$

Consequently

$$B_{\mathrm{new}} > k^2 q^2 (Y^2/4 - q^6) > 0.$$

Retain the arithmetic coordinates, set $v = Y$ and $n = t - g - v_{\mathrm{scale}}$, and represent B_{new} by the three Pos coordinates. Then $Z/U = t$, and all four new rows vanish. These constructions prove equivalence of zero existence; a bijection of arbitrary zero sets is not asserted.

6.7 Exact degree, reproducibility, and boundary

The four residual degrees are $22, 12, 16, 18$. In degree 22, the first residual contributes only $4ULZ$. With subscript 0 for leading homogeneous forms,

$$q_0 = 6\beta j, \quad X_0 = 12q_0 w, \quad Y_0 = v,$$
$$R_0 = -q_0^3(12wC - q_0 z), \quad U_0 = 8q_0^3 X_0 Y_0, \quad L_0 = X_0 Y_0^2 R_0,$$
$$Z_0 = U_0 n + 4q_0^3 X_0^2, \qquad P_{44} = 16U_0^2 L_0^2 Z_0^2.$$

In particular,

$$[j^{26}w^6 v^8 z^2 n^2]P = 2^{54}3^{32}\beta^{26} > 0.$$

The exponents sum to 44, and the other squared rows have smaller degree. The exact degree therefore holds after fixing every coefficient.

The accompanying independent cut-register audit authenticates coefficient bytes, checks the literal graph, exact identities and leading coefficient, and rejects altered arithmetic gates. These finite checks validate the transcription and algebra, not the unbounded inherited compiler theorem. The result is an ordinary arithmetic equivalence, hence an equivalence provable in ordinary ZFC between no integer zeros and syntactic Con(ZFC). It is not a proof of consistency, a global optimum, or a new formal-kernel certificate.

7 Complete formulas for the positive polynomial family

To make the next section independent of a reader opening the predecessor, we give the literal polynomial definitions here. The ten unknowns

$$J, F, \alpha, i, T, x, Y, n, \eta, \zeta$$

are strictly positive integers. The four fixed ports are the original compact44 tuple, as distinguished above. Put

$$q = \beta J + 1, \quad X = q + x, \quad \mu = q^2 + 1, \quad p = \mu Y,$$
$$A = p(X + 1) + 1, \quad \Delta = A^2 - 1, \quad H = 4A - 5,$$
$$k = \eta + \zeta, \quad c = kY + \eta, \quad v_0 = p^2 X,$$
$$D = v_0(v_0 + 1)k^2 + 1, \quad E_0 = ic^2, \quad E = 2E_0,$$
$$G = (q - 1)(q - F) + \alpha,$$
$$R = G(q^2 - 1) + (M_C + qM_F)J, \quad e = k - R - 1,$$
$$C = q - F - \alpha, \quad B = (qK_0 + X)C - qF,$$
$$F_s = 1 + \Delta DE^2, \quad Q = \Delta^2 DE^2, \quad \delta = Q - 1,$$
$$b = c(T + \alpha) - R, \quad V = F_s b - c, \quad D_{\mathrm{num}} = QV^2 - 1,$$
$$P = q^4(q - 1), \quad M = P\mu^2 H, \quad K_{\mathrm{pos}} = X(3c - 2X) + 2.$$

For $j = 1, 2$, use

$$Q_1 = pv_0 k^2, \qquad Q_2 = p(v_0 - p^2)k^2, \qquad S_j = ADE - E_0 Q_j.$$

Then define

$$Z_j = P\mu^2[H(n + V + S_j - bB) + K_{\mathrm{pos}}]$$
$$+ H[\mu^2(q(q - 1)Y - q^3 B - (q - 1)X) - Pe],$$
$$s_j = \delta Z_j^2, \qquad u = \delta M^2 DF_s, \qquad v = M^2 D_{\mathrm{num}}.$$

The sole equation for each sibling is

$$\boxed{N_j = (s_j - u - v)^2 - 4uv = 0.}$$

These expressions are exactly the formulas in `positive/PROOF.md`; the graph files `first.json` and `second.json` specify shared evaluation schedules, not a different polynomial. In those graphs, `I` denotes E, and neither symbol introduces an unknown.

8 Polynomial radical centering: ten positive unknowns at degree 252

The new circuits have respectively $(102, 253)$ and $(104, 252)$ as (live binary operation count, exact total degree). They retain ten strictly positive unknowns and the complete original compact44 coefficient tuple. This is a new degree–cost tradeoff, not a nine-variable result, a new Lean endpoint, or a proof that ZFC is consistent.

Keep all definitions of the retained auxiliary-majorant uncentered construction, except set

$$p = \mu Y, \qquad E_0 = ic^2, \qquad E = 2E_0.$$

Thus $A = p(X+1)+1$, $v_0 = p^2X$, $D = v_0(v_0+1)k^2+1$, $F_s = 1+(A^2-1)DE^2$ and $Q = (A^2-1)^2DE^2$. Define

$$Q_1 = pv_0k^2, \qquad Q_2 = p(v_0 - p^2)k^2, \qquad S_j = ADE - E_0Q_j.$$

Replace n by $n + S_j$ in the uncentered carrier, keeping the same quartic norm. Both resulting expressions are integer polynomials, fully specified by `first.json` and `second.json`.

Whole-zero-set equivalence. At every integer input there is the polynomial identity

$$N_j(\ldots, i, \ldots, n, \ldots) = N_{\mathrm{old}}(\ldots, 2i, \ldots, n + S_j, \ldots).$$

For positive inputs, $0 < Q_2 < Q_1 < D$ and hence $0 < S_j < ADE$. The transformed old coordinates are therefore positive without any Pell, parity or transport assumptions. The retained old soundness theorem applies directly. For completeness, choose the strong Pell recurrence period modulo $2\tau c^2$ rather than τc^2, obtaining $I = \tau E = 2\tau ic^2$. The old explicit auxiliary witness has $h_y = y - V > 9Q/4$. The canonical integer carrier is

$$n = Wt + \rho + \tau f - S_j + h_y - \kappa - s_Y + u_X.$$

Here $t, \rho, u_X \geq 0$, $\kappa, s_Y < I$, and $S_j - \tau f < ADE - \tau f < DE < I^2$. Thus $n > 0$, including $t = 0$. The carrier equals $\tau f + y$, so a norm factor vanishes. This proves both directions relative to the unchanged retained source61/native/controller calibration.

Exact degrees. For the formal Laurent root $g = \sqrt{DF_s}$ near ADE,

$$ADE - g = \frac{DE}{2A} + O_{\deg}(9), \qquad S_j - g = \frac{E_0(D - AQ_j)}{A} + O_{\deg}(9).$$

The exact remainders are

$$D - AQ_1 = v_0(1 - p^2 - p)k^2 + 1,$$
$$D - AQ_2 = (p^4 - p^3X + p^3 + p^2X)k^2 + 1.$$

They have degrees 15 and 14; therefore the centered defects have degrees 16 and 15. In the exact four-factor norm, the other three factors have degrees 38, 38 and 25, while $\deg(\delta^2M^4) = 2(42) + 4(13) = 136$.

The resulting exact degrees are $136+38+38+25+16=253$ and $136+38+38+25+15=252$. With $q_0=\beta J$, $X_0=q_0+x$ and $k_0=\eta+\zeta$, the highest forms are

$$[N_1]_{253}=-2^{20}q_0^{122}X_0^{27}Y^{60}i^{10}k_0^{32}(q_0-F)^2,$$
$$[N_2]_{252}=2^{20}q_0^{122}X_0^{26}Y^{60}i^{10}k_0^{32}(q_0-F)^2.$$

These are nonzero at the authentic fixed ports. The full proof and reproducible normal/optimized Python receipts are in the accompanying packet. Finite controls check transcription and algebra, not the unbounded calibration theorem.

9 Eleven integers, degree 352: saturating by a hidden square root

The primary equation below has eleven unrestricted integer unknowns, exact total degree 352, and 132 live binary arithmetic operations. It improves the incoming degree-416 endpoint in degree, at a cost of seven operations. Three new nondominated companions have degree/cost pairs 428/123, 376/127, and 364/131. The six fixed source34 numerals are unchanged. This is a new ordinary arithmetic reduction with the inherited native/compiler/refutation calibration, not a consistency proof, a global minimum, or a new end-to-end proof-assistant certificate.

9.1 The complete polynomial

The unknowns are

$$(j,C,z,h,r,v,w,T,c,n,s)\in\mathbb{Z}^{11}.$$

Use the fully authenticated numerals $\beta,K_0,M_C,M_F,A_c,B_c$, with $\beta=2^{1064921}-1$, $A_c=2^{1113}$, $B_c=2^{666104}$ and the already shifted M_F. Every symbol below is a polynomial abbreviation:

$$J=6j+1,\quad q=\beta J+1,\quad W=12w+4,\quad X=qW,\quad Y=v,$$
$$E=XY,\quad a=E+Y,\quad A=a+2,\quad H=4a+3,\quad \Delta=a^2+H,$$
$$F=(K_0+W)C-z(q-1),$$
$$R=(q(q-F)-C)(q^2-1)+(M_C+qM_F)J,$$
$$k=R+1+hE,\quad \eta=c-Yk,$$
$$D_{\mathrm{err}}=(A_cF-q)^2+(B_cC-q)^2,$$
$$B_0=k^2(q^2-D_{\mathrm{err}})-4q^2\eta^2,\quad \mathcal{B}=Y^2B_0-k^2q^8,$$
$$\ell=XY^2,\quad L=\ell k,\quad D_\tau=1+L(L+k),$$
$$D=3\mathcal{B}-4,\quad D_g=1+4DD_\tau s^2,\quad P=D_\tau D_g,$$
$$D_m=1+\Delta c^2,\quad U_b=rc^2,\quad U=AU_b,$$
$$D_f=1+\Delta U^2D_m,\quad Q=\Delta^2U^2D_m,\quad \delta=Q-1,$$
$$V=D_f(c-R+4UD_mT)-c,\quad C_y=QV^2-1,\quad D_a=\delta C_y,$$
$$K=X(3c-2X)+2,\quad M=2q^3,\quad D_c=HM,\quad S=srHM^5,$$
$$z_c=D_cT+MK+H(Y-q^3),$$
$$\mathcal{A}=(z_c^2+3D_c^2)(z_c^3-19D_c^3),$$
$$\lambda=(2\Delta+1)U_bD_m,\quad Z=S(n+\lambda V)-\mathcal{A}.$$

The final equation is the ordinary integer polynomial

$$\boxed{\left(\delta Z^2+\delta S^2P-4S^2D_mD_fC_y\right)^2-4\delta^2S^2PZ^2=0.} \tag{8}$$

Denote the left-hand side of (8) by Φ_{352}. Its literal evaluator is `integer/free_main_saturated11_sharp.json`. It has no division, radicals, variable exponents, or hidden witnesses. The displayed radicals in the proof are only an analysis of this polynomial.

9.2 Why one main square root can replace a full factor

Unconditionally, q is even, $q \equiv 2 \pmod 3$, $|q| \geq \beta - 1$, $8 \mid X$, $X \equiv 2 \pmod 3$, and $|X| \geq 4|q|$. The integer H is odd and nonzero. If $Y = 0$, then $\Delta = 3$; otherwise $|a| \geq |X| - 1$, so in either case $\Delta = A^2 - 1 > 1$. Thus $D_m, D_f > 0$. Also $4 \mid Q$, $D_f \equiv 1 \pmod 4$, δ is odd and nonzero, and D_τ, P are odd, with $D_\tau \geq 1$. These parity assertions do not assume Y even: ΔA is always even.

The intersective quintic $\pi(t) = (t^2 + 3)(t^3 - 19)$ has no rational root, so $\mathcal{A} = D_c^5 \pi(z_c/D_c) \neq 0$. If $S = 0$, the output is $\delta^2 \mathcal{A}^4 \neq 0$. Every zero therefore has $r, s \neq 0$. If $c = 0$, then $D_a = 1$. If $c \neq 0$, then $|U| \geq 2c^2$ and $D_f > |c| + 1$; the integer-distance argument in $V = D_f b - c$ gives $V \neq 0$, $Q > 1$, and $D_a > 0$.

Equation (8) is δ^2 times the product of

$$Z \pm 2S\sqrt{D_m D_f C_y/\delta} \pm S\sqrt{P}.$$

The first radicand equals $D_m D_f D_a/\delta^2 > 0$. Thus P cannot be negative at a zero: it would give the sole nonzero imaginary term in a vanishing factor. Hence $D_g > 0$ and $\mathcal{B} \geq 2$. The stronger budget yields

$$k \neq 0, \quad B_0 > 0, \quad |Y| > |q|^3, \quad D_{\mathrm{err}} < q^2, \quad |c/k - Y| < \tfrac{1}{2}.$$

It supplies the authentic common-sign packing bounds and excludes $c = 0$.

Multiplying a normalized factor by δ gives a rational signed sum of $2\sqrt{D_m D_f D_a}$ and $\delta\sqrt{P}$. Since P and δ are odd, the two terms have different 2-adic orders if they belong to the same nontrivial squareclass; they cannot cancel. Distinct squareclasses are linearly independent. Thus both $D_m D_f D_a$ and P are squares.

Crucially, D_m, D_f, D_a are pairwise coprime. Modulo D_m, $D_f = D_a = 1$. Modulo D_f,

$$Q = -\Delta, \quad \delta = -A^2, \quad V = -c, \quad C_y = -D_m, \quad D_a = A^2 D_m.$$

Because $A \mid U$, D_f is coprime to A as well as D_m. Consequently

$$d^2 = D_m, \quad f^2 = D_f, \quad e^2 = D_a, \quad \tau^2 = D_\tau, \quad g^2 = D_g$$

for positive integers; the last split uses $\gcd(D_\tau, D_g) = 1$. The new integer coordinates used only in the proof are

$$I = Ud, \qquad T_{\mathrm{old}} = dT.$$

They satisfy $c^2 \mid I$ and restore exactly

$$f^2 - \Delta I^2 = 1, \quad Q = (\Delta I)^2, \quad V = f^2(c - R + 4IT_{\mathrm{old}}) - c.$$

The rational $y = e/\delta$ is integral by adjacent-Pell denominator descent: its norm-preserving transformation decreases $|V|$ while preserving a reduced denominator, ending at $|V| = |y| = 1$. The factor equation now gives $S \mid \mathcal{A}$.

9.3 Late scale recovery without early coprimality

The divisibility $M^5 \mid \mathcal{A}$ forces $M \mid z_c$ by the unique lowest valuation of z_c^5 when $v_p(z_c) < v_p(M)$. Hence $M \mid H(Y - q^3)$. Its 2-primary part proves Y even, but we do not yet cancel H modulo q. Also $H \mid \mathcal{A}$ gives $H \mid (MK)^5$. With

$$K_q = X(X + 2ac) - 1, \qquad K = HXc - 2K_q,$$

oddness of H gives $H \mid (q^3 K_q)^5$.

The retained signed pre-projection rank argument now has every required hypothesis: the cubic-scale magnitude, centered strip, common signs, first and main norms, $c^2 \mid I$, and the two exact auxiliary residues $V \equiv -c \pmod f$, $V \equiv -R \pmod c$. Its nonodd strong-rank step gives $c \mid m$ for the strong Pell index; it does not assume that I/c^2 is odd. It yields

$$R = 2t + 1, \quad t \geq 36, \quad c = \operatorname{sign}(c)\psi_A(R), \quad |H| > |X|(|X| - 3)^t.$$

Here $t \geq 36$ follows already from $R > |q|(|q|^2 - 1) \geq 4080$.

Put $x = |X|$, $\sigma = \operatorname{sign}(c)$ and $\Pi = (\sigma X - 2^R)(2^R \sigma X + 1)$. The Pell projector congruence implies $H \mid q^{15}\Pi^5$. But

$$|q|^{15}|\Pi|^5 < 2^{20t-10}x^{25}, \qquad |H| > x^{t+1}/2^t.$$

The unconditional $x \geq 4(\beta - 1) \geq 2^{64}$ and $t \geq 36$ give $64(t - 24) - (21t - 10) = 43t - 1526 \geq 22 > 0$. Thus the divisible integer is smaller than $|H|$ and must be zero. Its odd second factor is nonzero, so $\sigma X = 2^R$. Now $q \mid X$ makes $|q|$ a power of two. Since H is odd, $\gcd(H, M) = 1$ finally holds, yielding

$$Y = (2v_{\text{old}} + 1)q^3.$$

The source's orientation, centered half-binomial interval, and fresh odd-kernel restoration apply. The full proof explicitly constructs the fresh strong/auxiliary witnesses; it does not relabel incoming I/c^2 as odd. An actual old signed Q41 zero is then passed to the retained native/compiler/refutation theorem.

9.4 Completeness with one fixed strong pair

Start with a canonical native witness. It has $D_m = d^2$, $D_\tau = \tau^2$, $B_0 > q^2k^2/4$ and $Y > X = 2^R > 4q^3$, so the stronger budget exceeds 64. Since DD_τ is positive and 2 modulo 3, ordinary Pell existence and squaring a unit give the required $s, g > 0$. Choose one strong pair with

$$4sAdc^2 \mid I, \qquad f^2 - \Delta I^2 = 1, \qquad r = I/(Adc^2).$$

All primes of dsr divide this fixed I. The retained supported-modulus interpolation theorem realizes every T_{old} class modulo dsr without changing this strong pair. Write $\kappa = K/H$. Choose an intersective root a_0 of $\pi(T + \kappa + v_{\text{old}})$ modulo sr, and prescribe

$$T_{\text{old}} = da_0 \pmod{dsr}.$$

Then $T = T_{\text{old}}/d$ is integral and $\mathcal{A}/S = H^4\pi(T + \kappa + v_{\text{old}})/(sr)$ is integral. Finally set

$$n = \mathcal{A}/S + 2dfy + \tau g - \lambda V.$$

A conjugate factor vanishes. The order is outer data, guard, fixed strong pair, intersective residue, auxiliary interpolation, and n. This proves both directions of the ordinary zero-existence reduction.

9.5 Exact degree, evaluation, and verification

The important degrees are

$$\deg(S, D_m, D_f, Q, V, C_y, \lambda, \mathcal{A}) = (20, 8, 26, 32, 41, 114, 17, 35).$$

Writing $W_0 = Sn - \mathcal{A}$, the sharp identity

$$\lambda^2 - 4D_m D_f = D_m^2 U_b^2 - 4D_m$$

shows that the inner bracket has unique leading term $S^2 Q D_m^2 U_b^2 V^2$, of degree 176. The final correction has degree 316, so the exact output degree is 352. An attained monomial is

$$[j^{124} w^{64} v^{64} s^4 r^{24} c^{68} T^4]\Phi_{352} = 2^{288} 3^{188} \beta^{124} > 0.$$

The graph shares the exact expressions

$$\mathcal{B} = (Yk)^2(q^2 - D_{\mathrm{err}}) - q^2(4(Y\eta)^2 + (kq^3)^2), \quad L = E(Yk), \qquad P = D_\tau + (12\mathcal{B} - 16)(D_\tau s)^2.$$

Every scalar multiplication and square is counted. The direct and simple centers give 376/127 and 364/131. A weaker-scale companion with $Y = qv$, the old budget and $M = 2q^2$ gives 428/123; its detailed early-CRT proof is in the package.

The local checker, without importing the builder, verifies every named register over free polynomial rings, all four conjugates, coprimality remainders, rescaling, exact cancellation cuts and attained gate leaders, topology/liveness, all six numeral payloads and their complete finite reconstruction, and rejects deliberate corruptions. It passes normally and with Python optimization. The retained source files and notices are hashed; their unbounded theorems are an explicit dependency, not something the finite checks claim to certify.

10 A single product improves both low-cost endpoints

The same main-radicand saturation yields a distinct low-cost pair:

$$(11 \text{ integers}, 924 \text{ degree}, 109 \text{ operations}), \qquad (11 \text{ integers}, 840 \text{ degree}, 110 \text{ operations}).$$

These strictly improve the incoming 966/111 and 882/112 pairs in both degree and evaluation cost. The degree-352 family is unchanged.

Use the same eleven variables and six actual coefficient payloads. Keep $J, q, W, X, F, R, a, A, H, \Delta$ from the ordinary source front, but restore the literal cubic scale and choose one of two index moduli:

$$Y = (2v+1)q^3, \quad E = XY, \quad k = R + 1 + hM_{\mathrm{index}}, \qquad M_{\mathrm{index}} = E \quad \text{or} \quad 4q^4.$$

Use the original centered budget, not the stronger free-scale one:

$$\eta = c - Yk, \qquad D_{\mathrm{err}} = (A_c F - q)^2 + (B_c C - q)^2,$$
$$\mathcal{B} = k^2(q^2 - D_{\mathrm{err}}) - 4q^2\eta^2,$$
$$L = XY^2 k, \qquad D_\tau = 1 + L(L+k), \qquad D = 3\mathcal{B} - 4,$$
$$D_g = 1 + 4DD_\tau s^2, \qquad P = D_\tau D_g, \qquad D_m = 1 + \Delta c^2.$$

The new definitions are

$$\begin{aligned}
B_P &= D_m P, & U &= Arc^2, & & \\
D_f &= 1 + \Delta U^2 B_P, & Q &= \Delta^2 U^2 B_P, & \delta &= Q - 1, \\
V &= D_f(c - R + 4UB_P T) - c, & C_y &= QV^2 - 1, & D_a &= \delta C_y, \\
K &= X(3c - 2X) + 2, & S &= srH, & z_c &= HT + K, \\
\mathcal{A} &= (z_c^2 + 3H^2)(z_c^3 - 19H^3), & Z &= Sn - \mathcal{A}. & &
\end{aligned}$$

The entire equation is now only

$$\boxed{\delta Z^2 - S^2 B_P D_f C_y = 0.} \tag{9}$$

Denote its left-hand side by Φ_{native} or Φ_{short}, according to the chosen index modulus. The literal files are `integer_lowops/product_main_guard11_native.json` and `integer_lowops/product_main_guard11_short.json`.

10.1 The sign branch and noncircular square recovery

Unconditionally, $D_m > 0$, $D_\tau \geq 1$, P is odd and nonzero, and δ, D_f, D_a are nonzero. The zero-free quintic makes $\mathcal{A} \neq 0$; at $S = 0$ the output is $\delta\mathcal{A}^2 \neq 0$. Thus $r, s \neq 0$ at any zero.

The three integers B_P, D_f, D_a are pairwise coprime, with no sign assumption: $D_f = D_a = 1$ modulo B_P, and modulo D_f one has $D_a = A^2 D_m$. Since $A \mid U$ and $D_m \mid B_P$, D_f is coprime to both factors of this remainder. Equation (9) gives

$$(\delta Z/S)^2 = B_P D_f D_a.$$

Their absolute values are therefore separately squares. But $\Delta A^2 = (A^2 - 1)A^2$ is divisible by 4, so $D_f \equiv 1 \pmod 4$. A negative square is impossible, proving $D_f > 0$. If $c \neq 0$, a negative B_P would make $D_f = 1 + \Delta U^2 B_P < 0$. If $c = 0$, then $D_m = D_f = D_a = 1$ and the product itself makes $B_P = P$ positive. Thus in all cases $B_P > 0$, $P > 0$, and $D_a > 0$. Write

$$p_0^2 = B_P, \qquad f^2 = D_f, \qquad e^2 = D_a.$$

Now $D_g > 0$ implies $\mathcal{B} \geq 2$. The centered strip excludes $c = 0$, because the literal Y is nonzero.

Set $I = Up_0$ and $T_{\text{old}} = p_0 T$. These restore the exact strong and auxiliary formulas; adjacent-Pell descent makes $y = e/\delta$ integral. At this point we deliberately recover the omitted main square *before* assuming the first-Pell square. The early absolute-folding theorem uses only

$$|A| \geq 3, \quad |c| \leq |I|, \quad f^2 - \Delta I^2 = 1, \quad V \equiv -c \pmod f, \quad QV^2 - (Q-1)y^2 = 1.$$

Classify $f = \chi_{|A|}(m)$, $|I| = \psi_{|A|}(m)$ and fold the odd auxiliary index modulo $2m$. The bound $\psi_{|A|}(p) + |c| \leq 2|I| < f$ prevents wraparound and gives $|c| = \psi_{|A|}(p)$. Hence $D_m = d^2$. Only now does $P = p_0^2/d^2$ become an integer square. Coprimality of D_τ, D_g then yields their separate square roots. This order avoids a circular invocation of the first-Pell classification.

Canceling δ gives $Z^2 = S^2 p_0^2 f^2 y^2$, so $S \mid \mathcal{A}$ and $H \mid K$[5]. The same nonodd strong-rank proof, fifth-power whole-product bound, orientation, half-binomial interval and fresh odd source restoration now apply. For the short modulus, $4q^4 > 3R + 1$ and $4q^4 \mid 2XY^2$ give the unchanged first-index no-wrap argument; after orientation its recurrence restores the literal native h. All dependencies and exceptional cases are stated in the complete `integer_lowops/PROOF.md`.

10.2 Completeness and exact metrics

Canonical source data give $D_m = d^2$, $D_\tau = \tau^2$ and $\mathcal{B} > 64$. First choose and fix the guard pair s, g by ordinary Pell existence, and set $p_0 = d\tau g$. Then choose one strong pair with

$$4sAp_0c^2 \mid I, \qquad f^2 - \Delta I^2 = 1, \qquad r = I/(Ap_0c^2).$$

Every prime of p_0sr divides this fixed I. With $\kappa = K/H$, choose a root a_0 of $\pi(T+\kappa)$ modulo sr, and use supported-modulus interpolation to require $T_{\text{old}} = p_0a_0 \pmod{p_0sr}$. Then $T = T_{\text{old}}/p_0$ is integral, $\mathcal{A}/S = H^4\pi(T+\kappa)/(sr)$ is integral, and $n = \mathcal{A}/S + p_0fy$ solves (9). The guard is fixed before the strong pair, and the strong pair is fixed before requesting an auxiliary residue.

The native degrees of $(P, B_P, D_f, Q, V, C_y, S, Z)$ are $(94, 108, 138, 150, 256, 662, 8, 35)$; the short degrees are $(82, 96, 126, 138, 232, 602, 8, 35)$. The second term of the output uniquely dominates, proving degrees 924 and 840. An attained native coefficient is

$$[j^{564}w^{106}v^{148}h^{42}s^{16}r^{12}c^{34}T^2]\Phi_{\text{native}} = 2^{960}3^{677}\beta^{564} > 0;$$

the short coefficient replaces $w^{106}v^{148}$ by $w^{64}v^{106}$ and is $2^{918}3^{635}\beta^{564} > 0$. Independent local checks authenticate the complete numeral recipes, bind every live register, prove the coprimality and rescaling identities, and certify exact degree and costs under normal and optimized Python. The finite audit is explicitly separate from the retained unbounded ordinary arithmetic and consistency calibration.

11 Independent verification and reproduction

The root program `reproduce.py` verifies the immutable inventory, the exact preceding release, and the hashes of the reviewed execution closure. It then makes isolated copies, deletes specified generated artifacts, rebuilds them, and compares exact bytes with the delivered files. The entire procedure is repeated under ordinary Python and Python with optimization enabled. Byte-identical receipts use Python 3.12.14 and SymPy 1.14.0; mode-specific receipts use the matching mode reference. Explicit exceptions, rather than removable `assert` statements, enforce the mathematical checks.

The direct checks reconstruct the literal cap formulas, all deleted and retained parent rows, the complete power-request coverage, the presemantic lower bounds, the signed Horn rewrite and each monic restoration. They collect the complete expanded polynomial, rather than infer a quartic degree or coefficient height from a circuit bound. The negative controls alter live cap rows, parent data, expansion coefficients and transformation certificates, requiring every alteration to be rejected.

The fifteen-integer audit checks every named register against an independent formula, every graph edge and live input, the exact highest homogeneous form and a nonzero degree-44 coefficient. It verifies the authentic coefficient bytes and rejects all 111 individual live-gate mutations. Its independent mathematical review addresses the amplified whole-product bound, the chronology of signed Pell recovery, late scale recovery, and constructive completeness. Numerical examples are not substituted for these arguments.

For the positive equations, an independent free-expression normalization proves whole-output equality with the old polynomial under the proposed coordinate substitution. The same review checks the full closed formulas, literal graph census, source bindings, both remainder identities, the complete four-factor norm and leading forms. The existence direction uses a fresh even Pell period and proves positivity of the carrier even at zero native transport.

The eleven-integer branch separates the local saturation and free-scale arguments from inherited native/compiler semantics. Its literal checks and independent review are distributed together with the exact six sibling graphs and the two separately reviewed low-cost product graphs. Degree bounds must be attained at the actual fixed coefficients; syntactic gate-degree bounds alone are not treated as exact degrees.

The archive also exercises missing-file, changed-equation, coefficient-byte, unsafe-path and ZIP-CRC controls. Its ZIP builder uses fixed metadata and requires two independent builds to be byte-identical. The root manifest is an integrity inventory, not a signature. None of these finite checks decides whether a refutation or polynomial zero exists.

11.1 Source closure and practical use

All current execution inputs are local. Historical source code is retained as inert evidence and is not run by the current replay. The exact previous release is kept under `prior/wave15`; its original manifest verifies every listed byte. The much larger working comparison snapshot is not bundled indiscriminately. The selected comparison evidence and its provenance are listed explicitly in `baseline/SELECTED_EVIDENCE.json`.

The new mathematical sections were copied from their producer sources, and report-specific additions are separately identified. The equations are preserved in TeX, not reconstructed from PDF text. The report is compiled with shell escape disabled, rendered, and visually reviewed page by page. Python replay and PDF review are separate checks with separate receipts.

12 Boundaries and useful next directions

No new nine-positive-variable or ten-unrestricted-integer construction is claimed. Nor does the 352 direct coordinate count compare universally with the separate native-coefficient quartic family. The nonnegative, positive and unrestricted domains continue to carry genuinely different constraints.

The gains here are concrete local transformations: a squared cap forcing signed indices together, an explicitly constructed odd auxiliary quotient, a free-scale magnitude guard followed by amplified projection, compatible main-root saturation, and polynomial approximation of a radical center. The exact equation, certificate and source pins make each transformation available for further work without conflating it with an overall simplicity record.

Further reciprocal terms in the positive center do not automatically remove the remaining reciprocal-X obstruction. A tempting rational center changes the congruence needed by the old source projection and is therefore not a proved endpoint. Similarly, a smaller universal carrier would require an unbounded existence argument with all simultaneous divisibility and nonvanishing requirements, not only successful modular experiments.

13 Provenance and source guide

1. Vladimir Reshetnikov, *ProveIt*, inspected comparison commit `f4d8db99e060`. The selected Git-blob and SHA-256 receipts are in `baseline/`. The late accepted comparison is `12a6f5a0376b`, 25 October 2026; its review is retained under `baseline/late/`.

2. *Paired Guards and Centered Carriers*, preceding delivered archive, 24 October 2026. Its exact extracted release and full inventory are retained under `prior/wave15`. The original ZIP SHA-256 is

   `773e07c4eb449b137febbb6714206a986`
   `53635caaf0c50cb86d00a99d8ce5458.`

3. Incoming *Shared Pell Bound 261*, natural pooled-cap theorem; incoming *Unified Binomial and Even Indices*, direct 442-integer and eleven-integer comparison points; incoming *Weak Carriers and Shared Signs*, degree-416 eleven-integer comparison point. These are credited source architectures, not new results of this packet.

4. The alternative branch's `PROOF.md`, source ledger, signed-cap review, and `inherited_proofs/` give the signed cap, odd quotient, direct arithmetic and C222 calibration chain.

5. `lowdegree/PROOF.md`, `SOURCE_PINS.json`, and `review/lowdegree/` give the free-scale carrier and independent mathematical review. `integer/PROOF.md` and its retained sources give the saturation sibling family.

6. `positive/PROOF.md`, `positive/prior/`, and `review/positive/` give the polynomial centers, o-riginal compact44 calibration and independent proof and literal-graph review.

Licenses and notices remain attached to their source closures. The root dependency guide distinguishes the current execution closure from historical evidence and the ordinary theorem assumptions that are not freshly mechanized.